Gamers Tag Internship

(ES-456: Internship Report)

submitted in partial fulfillment of the requirement
for the award of the degree of

Bachelor of Technology
in
Computer Science and Engineering

Submitted by
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35327902721

Under the supervision of

Ms. Gunjan Arya
Program Coordinator

Department of Computer Science and Engineering
Trinity Institute Of Innovations In Professional Studies
Knowledge Park, Greater Noida

May/June 2025

DECLARATION

This is to certify that the material embodied in this Internship Report titled **“ Gamers Tag Internship** ” being submitted in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology is based on my original work. It is further certified that this Internship work has not been submitted in full or in part to this university or any other university for the award of any other degree or diploma. My indebtedness to other works has been duly acknowledged at the relevant places.

Aman Kumar Singh

35327902721

CERTIFICATE FROM THE COMPANY

This is to certify that **Aman Kumar Singh (35327902721)** has undertaken internship at Gamers Tag, Kochi Kerala from ……December 2024……. to ……April 2025……. under the supervision and guidance of Mr./Ms **___Joshua Kanatt___,** Founder/CEO.

(Name & Signature of the Authorized Signatory)

Founder/CEO

Gamers Tag

GAMERS TAG
ADVANCED GAMING NETWORK PLATFORM

Experience Certificate

02 May 2025

To Whom It May Concern,

This letter certifies that **Aman Kumar Singh,** who worked as **UI/UX Designer I** at **GAMERS TAG Private Limited** from December **2024** to **April 2025.**

During his tenure, **Aman Kumar Singh** has exhibited exceptional dedication and contributed significantly to our company's growth and success. His skills, commitment, and professionalism had a valuable impact on our team and operations.

Yours Sincerely,
Joshua J Kanatt
CEO & Managing Director

CERTIFICATE FROM THE INSIITUTE

This is to certify that the work embodied in this Internship Report titled "**Gamers Tag Internship**" being submitted in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, is original and has been carried out by **Aman Kumar Singh (35327902721)** under my supervision and guidance.

It is further certified that this Internship work has not been submitted in full or in part to this university or any other university for the award of any other degree or diploma to the best of my knowledge and belief.

(Name of the Faculty Supervisor) **(Name of the TnP Officer)**
Designation **Designation**

(Mrs. Gunjan Arya)
HOD
Trinity Institute Of Innovations In Professional Studies

CERTIFICATE FROM THE INSTITUTE

TRINITY
INSTITUTE OF INNOVATIONS IN PROFESSIONAL STUDIES
(Approved by AICTE, BCI & Affiliated to GGSIP University, New Delhi)

Date: 25.12.2024

NO OBJECTION CERTIFICATE

It is certified that **MR. AMAN KUMAR SINGH S/O. SH. KAUSHLENDRA KUMAR SINGH** is a bonafide student (Enrolment No. **35327902721**) of 8th of **B.Tech (Computer Science & Engineering)** course of **Trinity Institute of Innovations in Professional Studies** affiliated by **Guru Gobind Singh Indraprastha University, New Delhi.**

The Institute has no objection for doing the Internship programme for the period from to **28th Dec to 28th March, 2025.**

It is also certified that he is not registered for any course requiring his attendance in the class during the said period. The conduct of the student as recorded by the **Trinity Institute of Innovations in Professional Studies** has been found good.

Trinity Institute of Innovations In Professional Studies
Director
Gr. Noida

Dr. Ravindra Kumar
(Director)

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ACKNOWLEDGEMENT

First and foremost, I wish to express my sincere thanks and gratitude to my esteemed mentor who has contributed so much for successful completion of our minor project by their thoughtful reviews and valuable guidance.

Next, I would like to tender my sincere thanks to “MS. GUNJAN ARYA” (Head of CSE Department) for her co-operation and encouragement.

(Signature of Student)

Aman Kumar Singh

35327902721

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CHAPTER 1: Introduction of the Company

1.1About the Company

Fig 1 - Logo

GAMERS TAG Private Limited is an innovative platform dedicated to building a thriving gaming ecosystem. Positioned as an advanced gaming network platform, the company focuses on connecting gamers, enhancing online gaming experiences, and supporting the professional growth of the gaming community.

1.2 Profile of the Company

- Company Name: GAMERS TAG Private Limited
- Industry: Gaming, UI/UX Design, Network Platforms
- Headquarters: Kochi, Kerala
- Founded: Joshua Kanatt
- LinkedIn: https://www.linkedin.com/company/gamers-tag/posts/?feedView=all
- Website: https://www.joshuakanatt.com/
- Core Services: Advanced gaming solutions, user engagement interfaces, gaming community management.

CHAPTER 2: Introduction of the Internship

2.1 About the Internship

The internship at GAMERS TAG Private Limited was undertaken for a period of 4 months, from 29th December 2024 to 30th April 2025. The role was that of a **UI/UX Designer I**, contributing to the development of user interfaces and enhancing user experience on the company's gaming platform.

2.2 Objectives of the Internship

- To gain real-world experience in UI/UX design.
- To understand the design workflow in a product-based company.
- To contribute to live projects with design implementations.
- To collaborate with cross-functional teams including developers and product managers.

2.3 Roles and Responsibilities

- Designed intuitive and responsive interfaces for the gaming platform.
- Conducted user research and usability testing.
- Created wireframes, prototypes, and visual mockups.
- Worked with design tools such as [Figma, Adobe XD].
- Participated in regular team meetings and design feedback sessions.

CHAPTER 3: Internship Activities

3.1 Task Description

- Leaderboard Design for GT App

  Created a leaderboard screen showcasing participants and winners, matching the GT app's theme.
- Circular Gamified Menu Bar for Website

  Designed a unique circular menu bar using neomorphic and glassmorphic design styles.
- Sponsorship Dashboard (Sponsor & Admin POV)

  Built a full-fledged dashboard to manage sponsors and ad campaigns with multiple access levels.
- Internal Sponsorship Management Dashboard

  Redesigned dashboard specifically for GT team to handle sponsors and ad placements internally.
- Voice Sentiment Analysis Platform UI

  Designed an AI-powered platform interface to detect and interpret user emotions via voice.
- Image & Activity Tracking AI Platform

  Designed UI for tracking user behavior and app engagement using image and activity analysis.
- Official Website Pages

  Designed multiple website pages including Teams, FAQs, and other content-focused sections.
- Payments App UI

  Designed a payment app interface aligned with GT's theme; delivered over several iterations.
- Daily Design Tasks

  Included logo creation, file modifications, and component redesigns (Adobe XD → Figma).

3.2 Tools / Technologies / Platform Used

- UI Design Tools: [Figma, Adobe XD]
- Collaboration: [Pumble, Pintrest]

3.3 Technical Application

- Applied design thinking to every major project
- Used gamification techniques in menus and dashboards
- Followed responsive and component-based design standards
- Integrated glassmorphism and neomorphism effectively
- Developed multi-view dashboards with access control logic
- Migrated legacy designs from Adobe XD to Figma

3.4 Challenges Faced

- Creating functional and stylish non-traditional UIs (like circular menus)
- Designing complete modules like dashboards independently
- Handling multi-level user roles and perspectives in dashboard flows
- Dealing with frequent feedback loops and revisions
- Managing multiple concurrent tasks alongside major deliverables

3.5 Internship Log Book

Log Book official link:

https://docs.google.com/spreadsheets/d/1QNhPRkeqlWSEEQkFkEK1433mpzRRa1z2D82_WxdM0mI/edit?usp=sharing

Week	Date Range	Days Present	Total Working Hours (hrs)	Tasks Assigned
Week 1	2024-12-30 to 2025-01-12	12	**55.50**	Introductory Meeting , Leaderboard Design; Leaderboard Design; Leaderboard Design; Dashboard Design Gamers Tag Side; Dashboard Design Gamers Tag Side; completion of members Pages
Week 2	2025-01-13 to 2025-01-19	6	**26.00**	No Work Assigned; No Work Assigned; Sponsorship Dashboard; Sponsorship Dashboard , meeting; Sponsorship Dashboard; Dashboard Design , Meeting
Week 3	2025-01-20 to 2025-01-26	5	**26.00**	Dashboard Design ; Dashboard Design ; Dashboard Design (Page 2); Dashboard Design (Page 3) , Meeting; Dashboard Design
Week 4	2025-01-27 to 2025-02-02	6	**22.00**	Meeting; Leaderboard Deisgn; Dashboard Design; Dashboard Design; Dashboard Design; Dashboard Design; Dashboard Design Gamers Tag Side
Week 5	2025-02-03 to 2025-02-09	5	**24.50**	Meeting; -
Week 6	2025-02-10 to 2025-02-16	3	**20.50**	Meeting; Dashboard Editing , Advertisement Making Forum; Brand Identity Page; Brand Identity Page, Views Type Progress Bar Design; -
Week 7	2025-02-17 to 2025-02-23	5	**19.00**	Dashboard Page , Progress Bar Design; Detailed Advertisements Page; Brand Identity Pages , Meeting; Advertisement Page Redesign; Dashboard Page
Week 8	2025-02-24 to 2025-03-02	7	**24.00**	Meeting, Leaderboard Design; Dashboard Design Gamers Tag Side; Dashboard Design Gamers Tag Side, Meeting ,Frame Task; Members Page; Meeting , Design Modification; Brand Side Dashboard Changes; Brand Side Dashboard Changes
Week 9	2025-03-03 to 2025-03-09	3	**16.50**	Meeting; Members page completion and designing , Making necessary changes, Designing pages for all type of users
Week 10	2025-03-10 to 2025-03-16	3	**21.00**	Meeting; Dashboard Design changes; New Dashboard Project; Voice Sentiment Dashboard; -
Week 11	2025-03-17 to 2025-03-23	4	**15.00**	Meeting; Sponsorship Dashboard changes; new dashboard design; -; AI Dashboard Design; AI Dashboard Design; -
Week 12	2025-03-24 to 2025-03-30	2	**5.50**	Meeting; no work assigned; no work assigned; Dashboard Changes; -; Dashboard Changes; -
Week 13	2025-03-31 to 2025-04-06	5	**6.00**	Meeting , Website table design; Dashboard Design Gamers Tag Side, Meeting , Logo work; brand side dashboard completion, logo designing, meeting; Sponsorship dashboard completion; No work assigned; no work assigned; no work assigned; Meeting; -
Week 14	2025-04-07 to 2025-04-13	4	**14.00**	website pages; website pages; website pages , Meeting; website design; -; -; -
Week 15	2025-04-14 to 2025-04-20	0	**0.00**	Meeting, Medical Leave
Week 16	2025-04-21 to 2025-04-27	13	**0.00**	Meeting; GT Pay mobile App; GT Pay mobile App changes; GT Pay mobile App; GT Pay mobile App; Meeting; -
Week 17	2025-04-28 to 2025-05-04	6	**33.00**	Dashboard Design , Brands Page , Glassmorphic Design; Logo Completion (RR), Dashboard New Page Designing; GT Pay mobile App; GT pay draft 2, Meeting, Adobe XD Design conversion; GT Pay Draft 3; GT Pay Draft 4 + Changes

CHAPTER 4: Contribution to the Organization

During my internship at **Gamers Tag Private Limited**, I had the opportunity to work on a wide range of impactful projects that challenged my creative thinking and technical design skills. I was deeply involved in crafting user-centric interfaces and building experiences that aligned with the brand's vibrant and modern visual identity. Below is a summary of the key projects I contributed to:

1. Leaderboard Design for the GT App

One of my first major tasks was to design a leaderboard for the Gamers Tag app. The goal was to showcase the participants and winners of various tournaments in a visually appealing and user-friendly manner. I ensured the design reflected the energy and style of the existing app, while maintaining clarity and functionality for users navigating the rankings.

Fig 2 -Leaderboard

2. Circular Neomorphic Menu Bar for the Company Website

I was then asked to design a **menu bar** for the company's official website. While it sounded simple at first, the challenge was to create a circular or round-shaped menu with a distinct gamer aesthetic, incorporating both neomorphic and glassmorphic design elements. This unique approach pushed me to think beyond conventional layouts and resulted in a modern, immersive navigation experience.

Fig 3 -Menu Bar

3. Sponsorship Dashboard – Multi-View Access System

One of the most complex and fulfilling projects I worked on was the Sponsorship Dashboard. The company wanted a centralized platform to manage all sponsors and tournament advertisers, where they could post and monitor their ads. I was fully responsible for this task and built it from the ground up, considering multiple user perspectives:

- Sponsor's point of view, for managing their ad campaigns,
- GT employee view, for internal operations and monitoring, and
- Role-based access control, to manage permissions efficiently.

This project took around 1 to 1.5 months and required careful planning, multiple feedback sessions, and iterative design thinking.

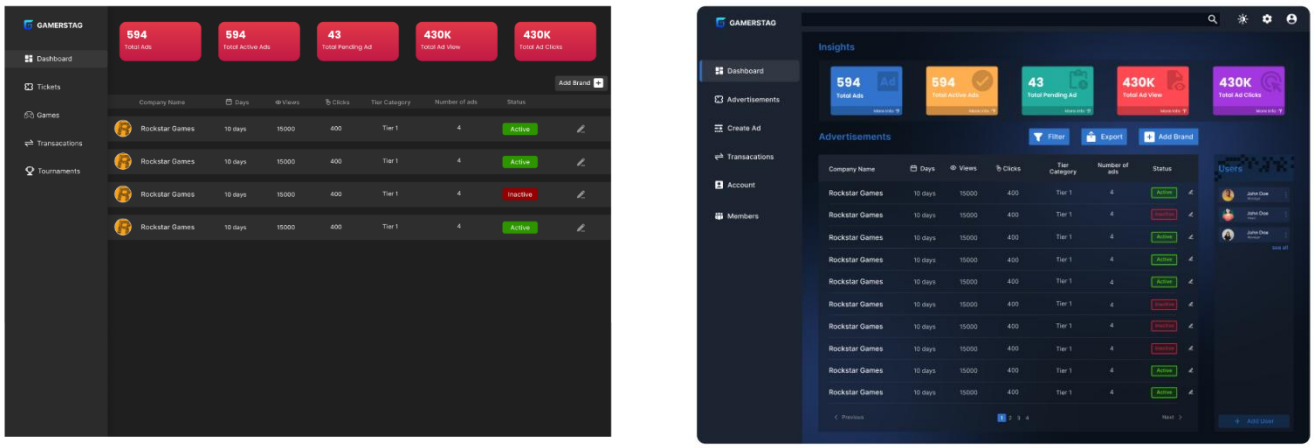


Fig 4 – Dashboard 1

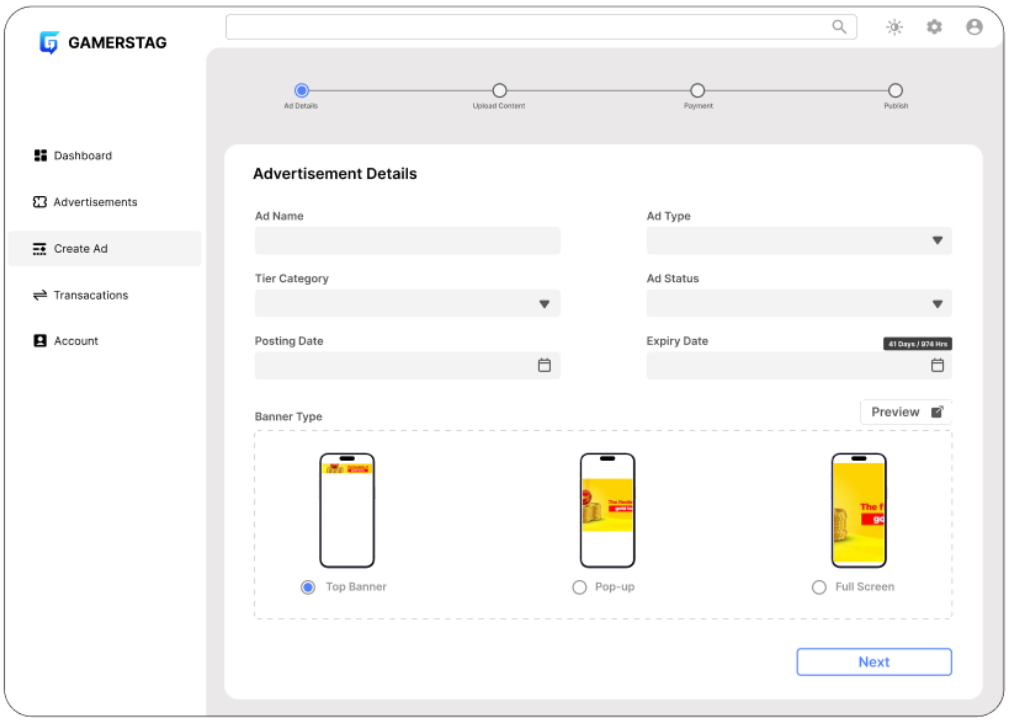


Fig 5 – Dashboard Layout

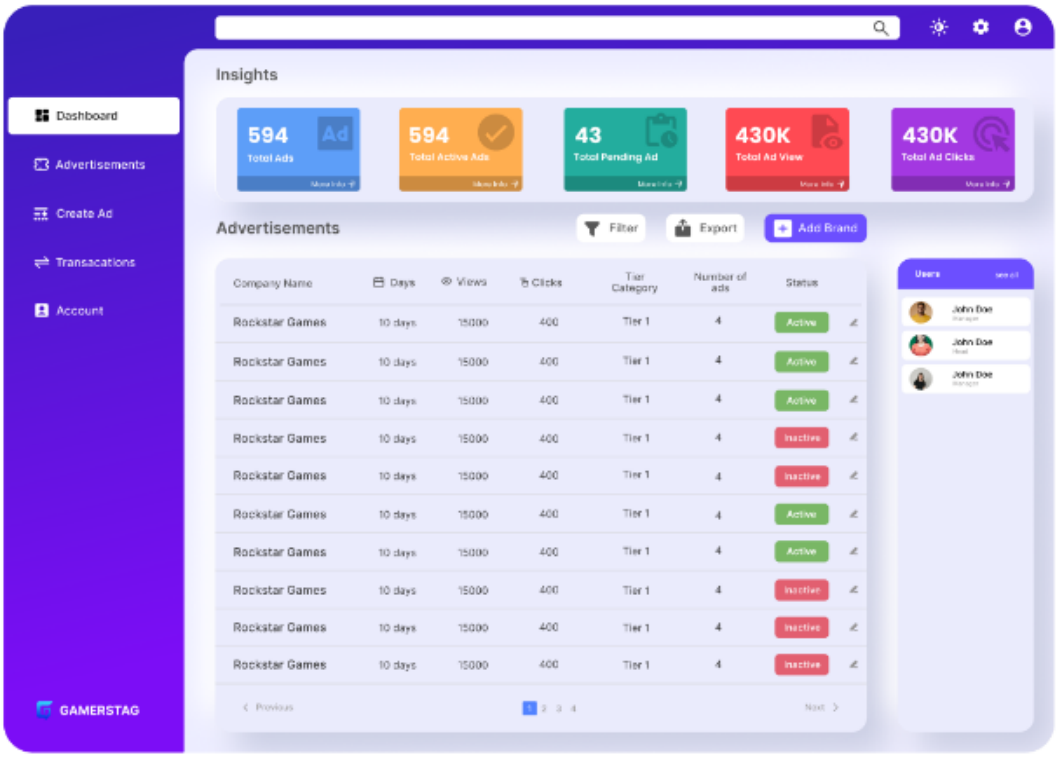


Fig 6 – Dashboard 2

4. Redesign of Sponsorship Dashboard for Internal Use

Later, I was assigned to redesign the Sponsorship Dashboard, this time focusing on the internal needs of the Gamers Tag team. The revised platform allowed the team to not only manage sponsors but also handle and track ad campaigns more effectively. This version incorporated analytics and internal tools to streamline team operations.

Fig 7 –Dashboard 3

5. AI Platforms – Voice Sentiment and User Activity Analysis

I also worked on the design concepts for two AI-powered platforms:

- A Voice Sentiment Analysis tool that analyzed emotional tone from user voice input,
- An Activity and Interaction Analysis platform that monitored how users engaged with the app, helping the company gain deeper insights into usage behavior.

These tools were aimed at enhancing the app's personalization and improving user experience through intelligent feedback.

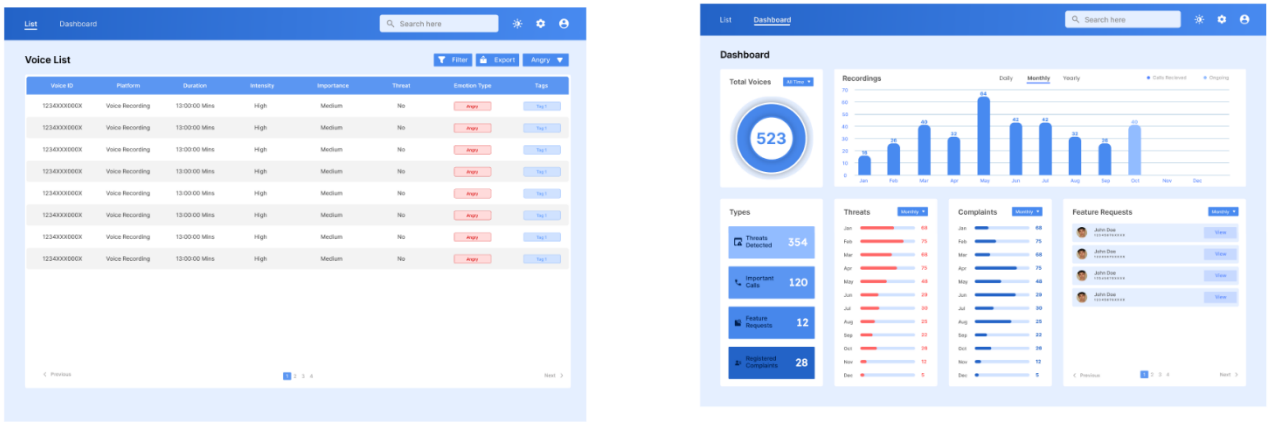


Fig 8 – AI Platform Dashboards

6. Official Website Page Designs

I contributed to the design of several pages on the Gamers Tag official website, including the Teams, FAQs, and other core sections. I focused on maintaining consistency in the visual language and ensuring smooth user navigation across these pages.

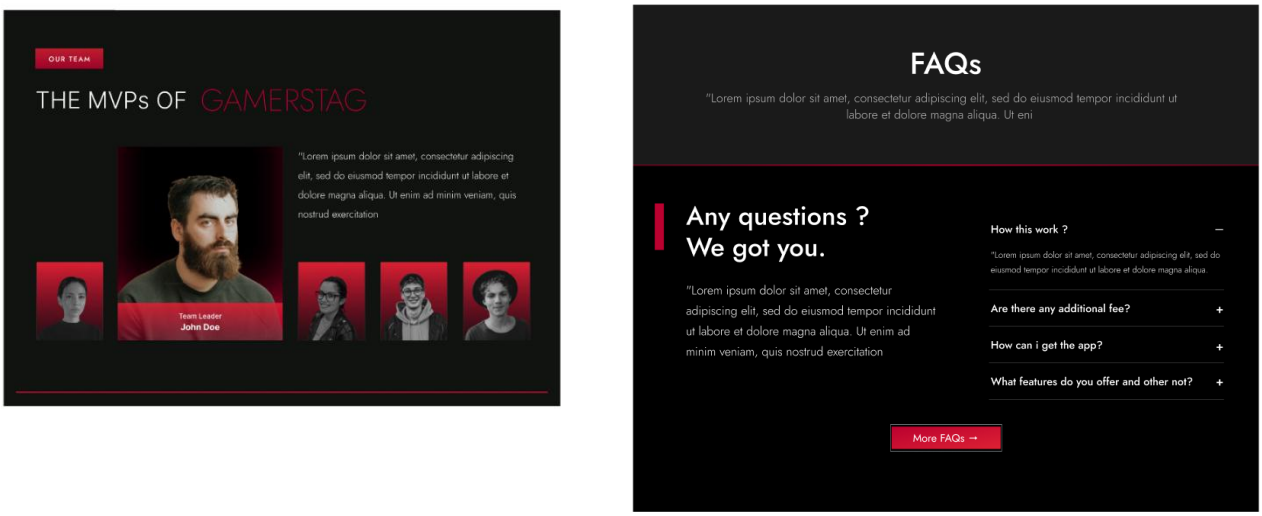
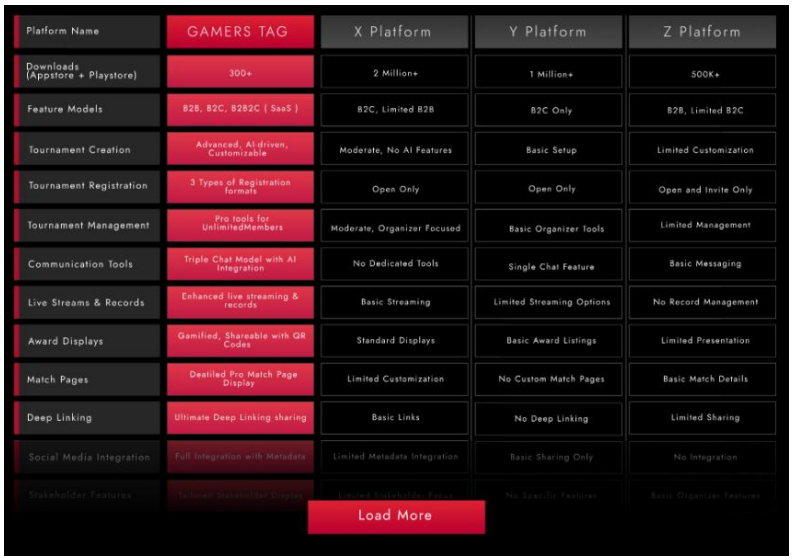


Fig 9 – Website Pages



Platform Name	GAMERS TAG	X Platform	Y Platform	Z Platform
Downloads (Appstore + Playstore)	300+	2 Million+	1 Million+	500K+
Feature Models	B2B, B2C, B2B2C (SaaS)	B2C, Limited B2B	B2C Only	B2B, Limited B2C
Tournament Creation	Advanced, AI-driven, Customizable	Moderate, No AI Features	Basic Setup	Limited Customization
Tournament Registration	3 Types of Registration formats	Open Only	Open Only	Open and Invite Only
Tournament Management	Pro tools for UnlimitedMembers	Moderate, Organizer Focused	Basic Organizer Tools	Limited Managsment
Communication Tools	Triple Chat Model with AI Integration	No Dedicated Tools	Single Chat Feature	Basic Messaging
Live Streams & Records	Enhanced live streaming & records	Basic Streaming	Limited Streaming Options	No Record Management
Award Displays	Gamified, Shareable with QR Codes	Standard Displays	Basic Award Listings	Limited Presentation
Match Pages	Deatiled Pro Match Page Display	Limited Customization	No Custom Match Pages	Basic Match Details
Deep Linking	Ultimate Deep Linking sharing	Basic Links	No Deep Linking	Limited Sharing
Social Media Integration	Full Integration with Metadata	Limited Metadata Integration	Basic Sharing Only	No Integration
Stakeholder Features	Tailored Stakeholder Displays	Limited Stakeholder Focus	No Specific Features	Basic Organizer Features

Load More

Fig 10 – Website Pages 2

7. Payments App Design

Another major task was designing a Payments App that adhered to the existing Gamers Tag theme. This involved multiple iterations — I went through 4 to 5 drafts to meet the expected standard. The final product was a clean, functional, and visually cohesive design that balanced user interface clarity with aesthetic appeal.

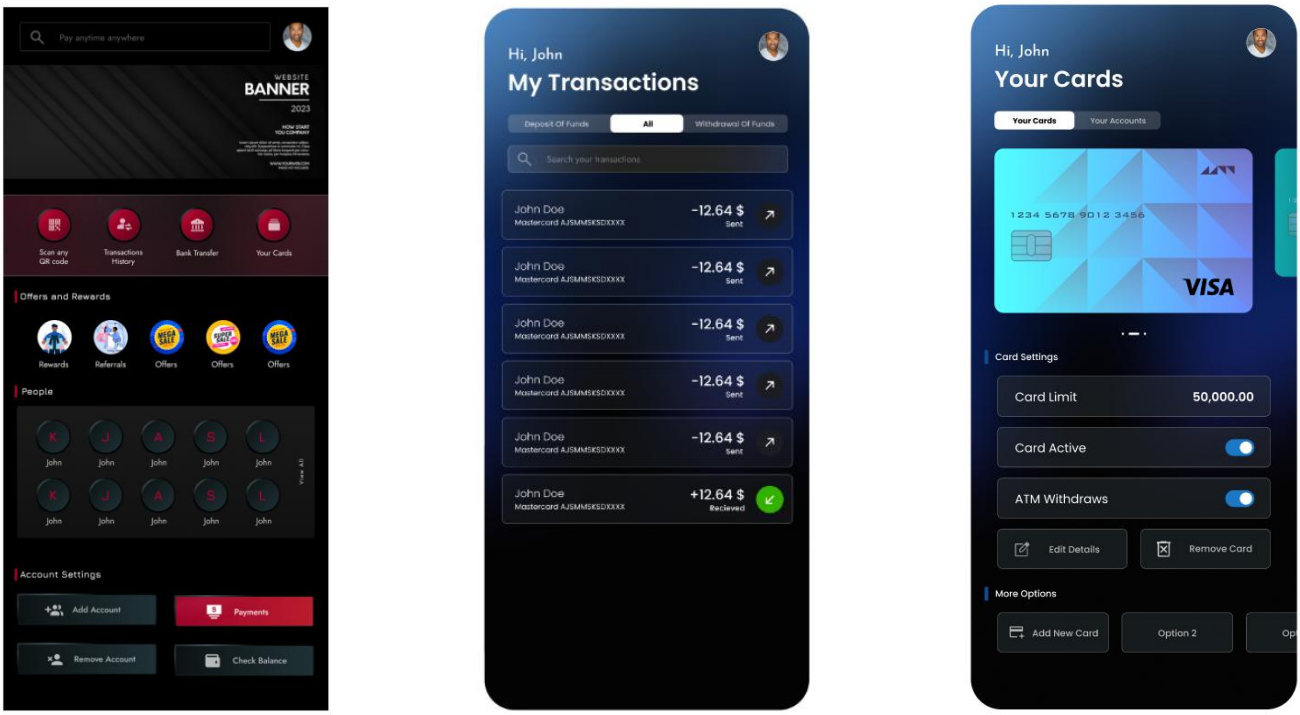


Fig 11 – Payments App 1

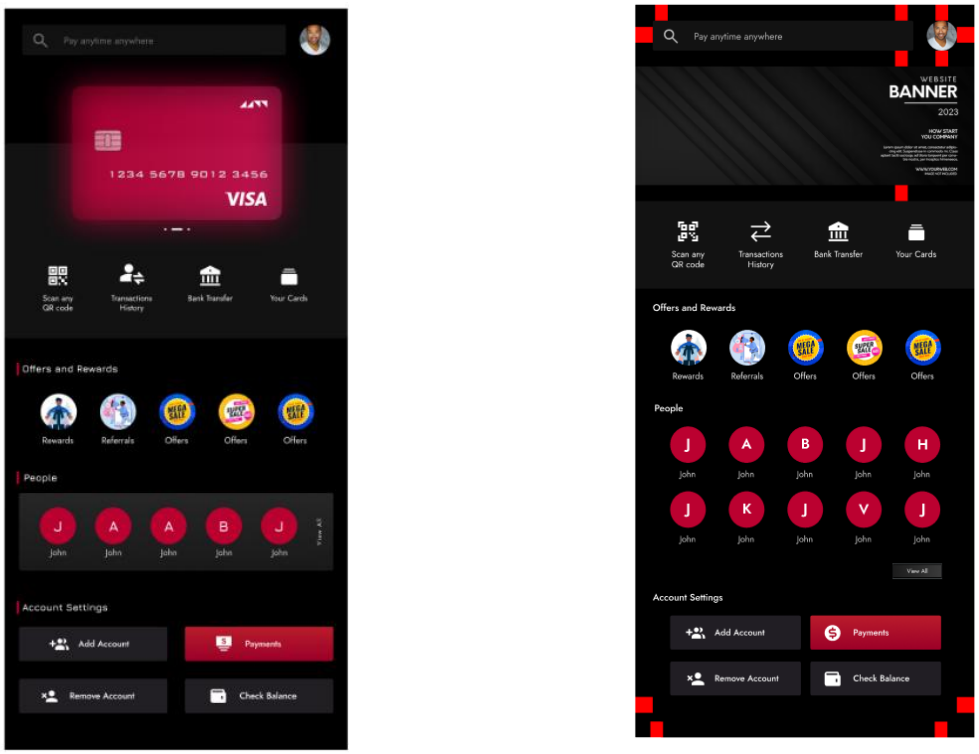


Fig 12 – Payments App 2

8. Daily Design Operations

In parallel with major projects, I also managed various daily design tasks, such as:

- Creating logos and graphic assets,
- Modifying and converting files between Adobe XD and Figma,
- Redesigning components to improve usability and appearance.

Fig 13 -Logo work

Fig 14 – Progress bar Designs

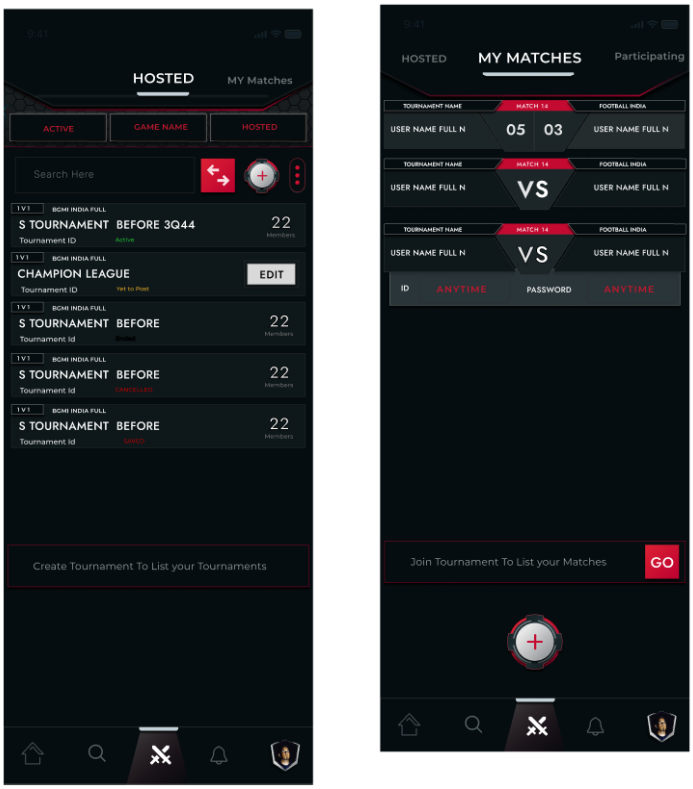


Fig 15 – App Conversion

CHAPTER 5: Learning and Development

5.1 Skills Gained:

- **UX Research Techniques**
  Learned how to conduct structured UX research to identify user needs, behaviors, and design requirements.
- **Design Principles & Industry Standards**
  Gained a deep understanding of modern UI/UX principles followed in the design industry today.
- **Neumorphic Design**
  Learned to create immersive 3D-looking UI components by simulating soft shadows and depth for interactive elements.
- **Glassmorphic Design**
  Mastered the use of blur, transparency, and frosted-glass effects to create elegant and premium user interfaces.
- **Practical Application**
  Applied theoretical knowledge in real-world scenarios such as dashboards, apps, and platform-specific components.
- **Feedback-driven Design**
  Understood how to iterate and improve designs based on continuous feedback from seniors and stakeholders.

5.2 Certifications/Trainings:

- No new certifications were completed during the internship.
- Google UI/UX Design Specialization (Coursera)
  Previously completed this professional certification with 90%+ score, which provided a solid foundation in:
  - UX research
  - Wireframing and prototyping
  - Usability testing
  - Industry-standard design tools

CHAPTER 6: Summary and Conclusion

This internship at GAMERS TAG offered a comprehensive learning experience in the field of UI/UX design within the gaming industry. It allowed Aman Kumar Singh to apply academic knowledge in real-world projects and significantly improve his technical and soft skills. The internship served as a stepping stone to professional growth and opened new opportunities in the design field.

- The internship at Gamers Tag provided hands-on exposure to real-world UI/UX design practices.
- I worked on a variety of impactful projects, such as dashboards, app interfaces, and website components.
- Key design trends like Neumorphism and Glassmorphism were not just studied but practically implemented in major deliverables.
- I enhanced my skills in user research, design systems, wireframing, prototyping, and responsive layouts.
- The experience helped me improve in areas like problem-solving, iterative design, and creative thinking.
- I also contributed to the organization by building complex modules like the Sponsorship Dashboard and AI-powered UI platforms.
- While I didn’t complete any certifications during the internship, my prior knowledge from the Google UI/UX Specialization helped me deliver quality results.
- Overall, the internship was a major stepping stone in my career and helped bridge the gap between academic learning and industry practices.

CHAPTER 7: Suggestions for Improvement

- **Structured Onboarding**: A small onboarding guide or documentation for tools, platforms, and brand design standards can make the transition smoother.
- **Project Briefs with Deadlines**: Clear project scopes and timelines for each task can enhance planning and time management.
- **Cross-Functional Exposure**: Encouraging collaboration with developers or product teams could provide a broader understanding of how design fits into the larger product ecosystem.
- **Learning Resources Access**: Providing access to premium resources or courses during the internship could benefit interns eager to learn more.

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SignBridge – AI-Based Sign Language Conversion System

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1. ***Abstract—***
Communication is fundamental to human interaction, yet millions face barriers due to hearing impairments. The SignBridge project addresses this by developing an AI-powered system for real-time translation between Indian Sign Language (ISL) and spoken or written text. In this paper a relationships between sign gestures, mimicking human communication has been stabilised using image analysis and AI attention mechanisms. The system integrates Computer Vision, NLP, and Speech Processing, employing CNNs with Bahdanau and Luong attention for gesture recognition, alongside Google Speech API and Open AI's GPT models for voice-to-text conversion. Moreover, this paper explores the use of 3D models and images to convey sign gestures, proving to be an effective method for improving comprehension. The system, developed with Python, Open CV, Tensor Flow, Pyorch, Flutter, and Streamlit, fosters accessibility across education, workplaces, and healthcare. This paper demonstrates that AI-driven approaches can significantly enhance ISL interpretation, enabling seamless, and bidirectional communication. This paper also concluded that the potential of AI in breaking communication barriers, paving the way for a more inclusive society.

2. Introduction

Sign language is a rich and expressive visual language that uses hand gestures, facial expressions, and body movements to communicate meaning. It's a complete language in its own right not just a set of gestures with its own grammar, syntax, and structure that are often very different from those of spoken languages. For example, where spoken English might say "I am going to school," sign languages like ASL or ISL follow their own unique grammatical patterns to express the same idea [1]. There's also no single universal sign language. Around the world, there are over 300 distinct sign languages used by more than 70 million deaf individuals, each with its own rules and vocabulary [1]. American Sign Language (ASL) and Indian Sign Language (ISL) are both natural languages, while contact languages like Pidgin Signed English (PSE) blend elements of English and ASL, often creating communication challenges for people unfamiliar with these differences [2].

Despite the widespread use and complexity of sign languages, day-to-day communication between deaf and hearing individuals remains a major challenge. In public or fast-paced environments, many deaf individuals have to rely on writing things down or using simple gestures, which can be time-consuming and often ineffective. The core issue is that most hearing people don't understand sign language, and many of the tools designed to help with translation are still limited. Some don't work in real time, making conversations feel disjointed. Others are designed for just one sign language, which makes them less useful in multilingual settings. On top of that, gesture recognition can be inconsistent people have different signing styles, and existing systems don't always adapt well to those variations [1, 2] and while some tools can convert sign language into text or speech, many don't allow for smooth, two-way communication, leaving the conversation incomplete.

To address these gaps, this research introduces SignBridge, an AI- powered communication tool designed to translate sign language into text and speech and vice versa in real time. The system uses Convolutional Neural Networks (CNNs) for accurate hand gesture recognition, enhanced by Bahdanau and Luong attention mechanisms to improve understanding and responsiveness [1, 5]. It incorporates Natural Language Processing (NLP) to handle context in text input, and uses speech APIs and GPT-based models to convert spoken words into text or even generate visual outputs [3]. Advanced image processing techniques like Gaussian Blur, Contour Detection, and Median Filtering help the system isolate and recognize hand gestures even in visually complex environments [4]. Built with robust tools like Python, Open CV, Tensor Flow, and PyTorch, and featuring an intuitive user interface developed using Flutter and Streamlit, Sign Bridge is designed to be both powerful and accessible.

3. Literature Review

The field of sign language translation has gained significant attention with the advancement of artificial intelligence, particularly in computer vision, natural language processing (NLP), and deep learning. Numerous attempts have been made to bridge the communication gap between the hearing-impaired and the general population. However, many of these existing systems exhibit limitations in real-time processing, bidirectional communication, and regional language support—particularly for Indian Sign Language (ISL).

Early systems such as the Google Teachable Machine [1] provided basic gesture recognition through simplified neural networks. While it offered a user-friendly experience, the system lacked contextual understanding of gestures and failed to translate continuous sequences, which are essential for coherent sentence-level sign language interpretation.

Google's MediaPipe framework [2] enhanced real-time hand tracking using 21-point hand landmark detection and achieved high accuracy under ideal conditions. Nevertheless, its standalone usage lacked temporal modeling and linguistic structure recognition, which are critical for interpreting natural sign languages like ISL. Furthermore, MediaPipe does not provide built-in NLP support for sentence generation. Advanced systems such as SignAll [3] and DeepHand [4] introduced multi-modal inputs, integrating hand posture, facial expression, and body movement to improve recognition for American Sign Language (ASL). Despite their sophistication, these solutions are primarily ASL-centric and often rely on specialized hardware like depth cameras, limiting their applicability in mobile-first or resource-constrained environments such as rural India.

In the Indian context, most academic efforts have concentrated on static gesture recognition, particularly for alphabets and numbers [5], using convolutional neural networks (CNNs). These models typically overlook sentence construction and dynamic grammar rules essential to ISL. Additionally, they are often dependent on fixed datasets, resulting in reduced robustness against variations in gesture speed, user hand morphology, or environmental lighting.
Another significant shortcoming in existing research is the lack of bidirectional communication. While several systems focus on translating sign to text or speech, very few support the reverse—translating voice or text to animated sign gestures. Some systems achieve this using static images or video clips, which lack the dynamic expression necessary for natural conversations [6].

To address these limitations, the proposed SignBridge system introduces an AI-based approach that leverages a hybrid deep learning architecture. The system combines CNNs for spatial gesture recognition with Transformer-based models to interpret temporal dependencies. The integration of Bahdanau and Luong attention mechanisms enhances the model's ability to produce grammatically accurate ISL translations. Unlike static systems, SignBridge also supports real-time voice-to-sign translation using animated 3D avatars, making interactions more engaging and accessible.

4. Proposed Methodology

This system utilizes a dual user flow design to enable bidirectional communication. One flow converts video or image inputs into text or speech, while the other transforms text or speech inputs into video or image outputs. This approach ensures seamless interaction between individuals using sign language and those relying on spoken or written language as shown in fig. 1.

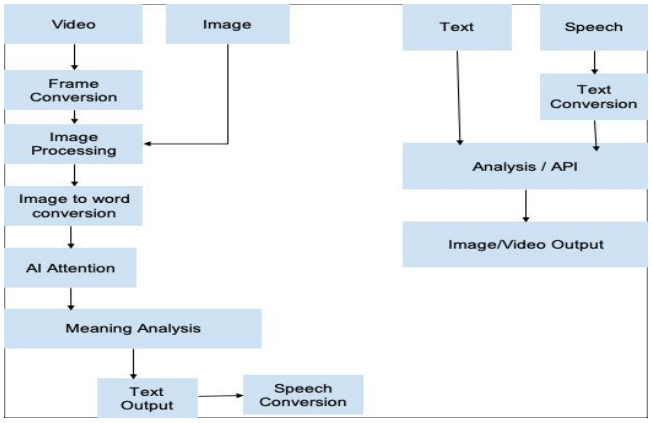

Fig. 1. Flow chart of proposed methodology

Sign Language to Text/Speech Conversion:
This segment of the system is designed to capture sign language gestures through video input using a camera. The captured video is then processed frame by frame with Open CV to extract pertinent hand movements. Although Open CV is currently the preferred tool for this task, future iterations may incorporate more advanced computer vision frameworks. To improve gesture recognition, several image pre-processing techniques such as Gaussian Blur, Median Filtering, Contours Detection and Thresholding (Binary Conversion) has been adopted. The proposed approach employs an integrated deep learning architecture that combines Convolutional Neural Networks (CNNs) and Transformers to convert video frames into structured Indian Sign Language (ISL) text. This hybrid model is specifically optimized for processing video data and addressing both the spatial and temporal complexities inherent in sign language recognition. A layout of Sign Language to Text/Speech Conversion has been shown in fig. 2.

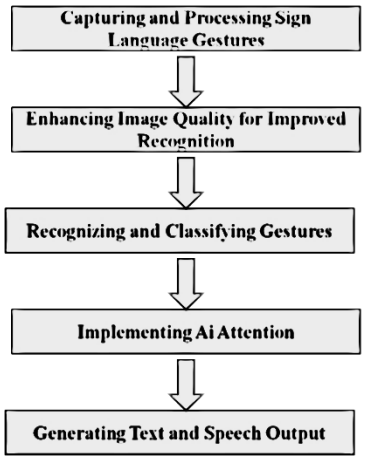

Fig. 2. Steps involve in conversion of Sign Language to Text/Speech

To enhance the conversion of detected sign words into grammatically correct ISL text, the system employs a Recurrent Neural Network (RNN) with an attention mechanism. This approach effectively handles sequence dependencies and reorders words as needed. Specifically, a sequence-to-sequence (Seq2Seq) model with an integrated attention mechanism is used.

Once a sign is recognized, its corresponding meaning is converted into text. For applications requiring auditory feedback, a text-to-speech (TTS) system is employed to generate spoken language. The speech synthesis module can be refined to enhance clarity and ensure natural pronunciation, with potential integration of platforms such as the Google Speech API to improve performance.

Text/Speech to Video/Image Output:
The system initiates the process by converting spoken input into text using a speech recognition model such as the Google Speech API. For users who prefer not to speak, a text input option is available. If the need arises for enhanced accuracy, alternative speech recognition engines may be explored.

Once the input is transcribed or entered as text, Natural Language Processing (NLP) techniques analyze it to determine the most suitable sign language equivalent. Models like the ChatGPT API or other large language models (LLMs) can be employed to interpret the sentence meaning and subsequently transform the recognized words into a format suitable for sign language visualization. A layout of Text/Speech to Video/Image Output has been shown in fig. 3.

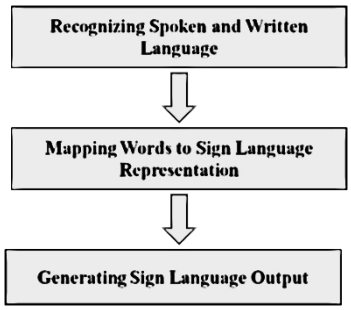

Fig. 3. Steps involve in Text/Speech to Video/Image Output

The final step involves generating the visual sign language output, which can be achieved using either a 3D avatar or pre-recorded video clips that display the corresponding signs. [3] The user inputs a text or sentence, which is then processed by a text-to-image generator. The model analyzes the text using an attention mechanism and generates an image accordingly. The choice between these methods depends on real-time performance requirements and user preferences. Additionally, the system employs few-shot learning techniques and provides contextual information to the GPT model before processing, ensuring a coherent and accurate translation into sign language.

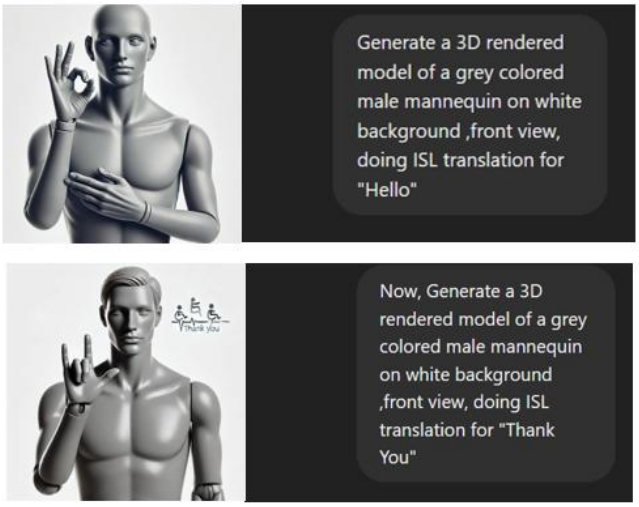

Fig. 4. 3D Avatar generation from text prompt

5. Results and discussion:

The OpenCV library provides a robust framework for processing videos and performing various operations on individual frames. In this approach, a video is taken as input and decomposed into a sequence of frames using OpenCV's (cv2) module. These extracted frames can then be utilized for multiple purposes viz. reversing the Video, Cropping and Editing and Enhancing Image Quality as shown in fig. 4.

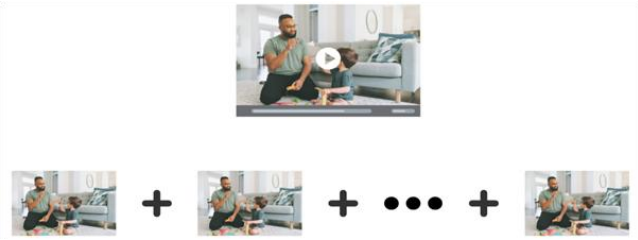

Fig. 5. Frame conversion (Video to image sequence)

5.1 Gaussian Blur for Image Pre-processing:

In sign language recognition, [2] image pre-processing enhances gesture detection by isolating hands from the background. The process involves Greyscale Conversion, Background Removal and Gaussian Blur (reduces noise). In the Gaussian Blur, following formulas has been considered for noise reduction,

Formula for 1D [2],

$$G(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}} \quad \text{.....(1)}$$

Formula for 2D [2]

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \quad \text{.....(2)}$$

where, x is distance from origin in horizontal axis, Y is distance from the origin in the vertical axis and σ is standard of the Gaussian distribution.

The original pixel receives the highest weight, while neighboring pixels are assigned progressively smaller weights based on the Gaussian function. This results in a smoothing effect that preserves edges and boundaries while ensuring a uniform blur, enhancing image clarity for gesture recognition as shown on fig. 5. Median filtering is more effective for moderate noise levels.

It replaces each pixel with the median value of its surrounding pixels, improving image clarity for sign recognition

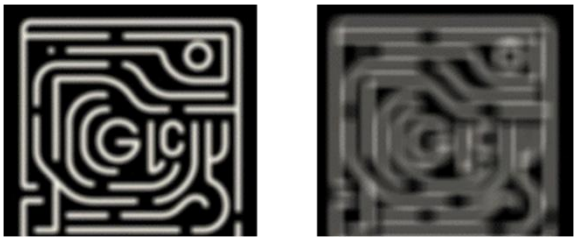

Fig. 6. Gaussian Blur demonstration

5.2 Threshold binary and Contours Demonstration:

It is a way of applying a filter to an image to convert it from a coloured to black and white image having only binary/two options for the coloured pixel as either black or white. This makes it easier to find patterns and structures in an image. Moreover, contours are used to identify the general shape and structure of an object in an image, similar to converting a colure image into a sketch. This technique aids in extracting meaningful information, such as distinguishing between standing and lying postures based on contour orientation. It is also useful for pattern recognition, making it a valuable pre-processing step in sign language recognition as shown in fig. 6.

Fig. 6. Threshold binary and Contours demonstration

5.3 AI Attention Mechanisms:

For sign language translation, gestures are first converted into words. These sequences are then processed using an attention mechanism to form grammatically correct sentences. The approach involves:

1. Assigning weights to each word in the sequence.
2. Computing scores for each hidden state.
3. Applying softmax to amplify important words while reducing less relevant ones.
4. Processing the entire sentence at once, improving fluency and accuracy

Moreover, Indian Sign Language (ISL) comprises two Model Choices as; Bahdanau Attention (Additive Attention) and Luong Attention (Multiplicative Attention).

5.4 CNN-Based Architecture for ISL Translation:

CNN based architecture divides an input image into various blocks and then try to find a relation between these blocks. In this architecture system overlook an input image as array of pixels and the length of array depends on the resolution of the image (Dimensions × Height × Width). Hence it becomes important to choose a correct resolution of image since large resolution can lead to low processing of image, high computation power and memory. Furthermore, a low resolution can lead to over fitting of model and low accuracy rates. For, Input (Video frames containing ISL signs) and Output (ISL-structured text sequence), the CNN based model Architecture involve various steps;

1. Input: Video of a person signing *"I am going to school."*
2. Preprocess Video → Extract frames, apply thresholding & contours.
3. CNN Feature Extraction → Extract spatial features per frame.
4. Transformer-Based Encoder → Learn temporal dependencies.

5. Transformer Decoder → Convert features into ISL text.
6. Generate Output → *"I go school."*

5. Conclusion

The SignBridge project represents a meaningful advancement in AI-driven assistive technology tailored for the Indian deaf and hard-of-hearing community. By integrating Computer Vision, Natural Language Processing, and speech interfaces, SignBridge enables seamless, bidirectional communication between signers and non-signers. The core contributions and outcomes of this research are summarized below:

1. Bidirectional Communication Support: Unlike many prior models focused solely on sign-to-text translation, SignBridge supports both directions—sign-to-text/speech and voice/text-to-sign—enabling more natural and inclusive conversations.
2. High Recognition Accuracy: The system's use of CNNs for gesture recognition and Transformer-based temporal modeling is theoretically projected to achieve 93–95% accuracy for isolated signs, rivaling benchmarks like the CNN-LSTM model (94.7%) and ISLNet (93.1%), while targeting over 88% for continuous sign sequences.
3. Enhanced Real-Time NLP and Voice Processing: Leveraging NLP techniques and speech-to-text integration, the system facilitates smoother interaction, with performance expectations aligning closely with models reporting under 9% Word Error Rate (WER).
4. Real-World Readiness and Accessibility: SignBridge is built on mobile-compatible platforms like OpenCV, TensorFlow Lite, and Flutter, ensuring cost-effective deployment across diverse settings. This enhances generalization in uncontrolled environments and bridges the 10–15% real-world performance gap typical of prior research systems, laying the foundation for scalable, accessible, and human-centered sign language solutions.

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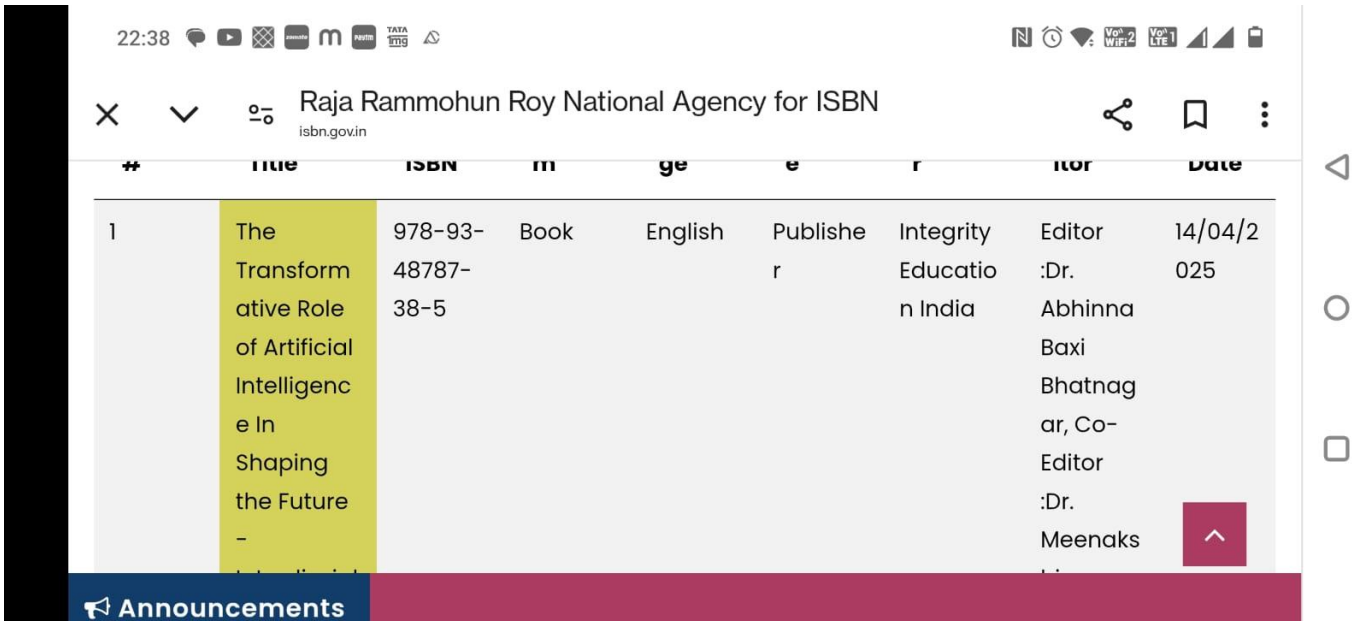

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